

An Introduction to $f(R)$ Gravity and Cosmological Applications

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Outline

- 1 Introduction & Motivation
- 2 Theoretical Framework
- 3 The Screened Logarithmic Model
- 4 Cosmological Dynamics
- 5 Conclusions & Outlook

The Dark Energy Problem

Observational Evidence:

- Type Ia Supernovae → accelerated expansion
- CMB anisotropies (Planck 2018)
- Baryon Acoustic Oscillations

Standard Λ CDM Model:

- Dark energy $\sim 68\%$ of total energy
- Cosmological constant Λ
- **Fine-tuning problem:** $\rho_{\Lambda}^{\text{QFT}} / \rho_{\Lambda}^{\text{obs}} \sim 10^{120}$

Alternative Perspective

“What if cosmic acceleration is not due to a new form of energy, but rather a signal that GR requires modification on cosmological scales?”

Modified Gravity: $f(R)$ Theories

From Einstein-Hilbert to $f(R)$:

$$S_{\text{EH}} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} R \quad \longrightarrow \quad S_{f(R)} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} f(R) + S_M$$

Why $f(R)$ gravity?

- Natural geometric extension of GR
- Self-accelerating solutions without explicit Λ
- Additional scalar degree of freedom: the **scalaron**
- Well-studied theoretical framework

Challenge

Viable models must satisfy stringent theoretical and observational constraints!

Field Equations in $f(R)$ Gravity

Modified Einstein equations:

$$f_R R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f_R + g_{\mu\nu} \square f_R = \kappa^2 T_{\mu\nu}$$

where $f_R \equiv df/dR$ and $\square \equiv g^{\mu\nu} \nabla_\mu \nabla_\nu$.

Trace equation:

$$f_R R - 2f(R) + 3\square f_R = \kappa^2 T$$

- In GR: $R = -\kappa^2 T$ (algebraic)
- In $f(R)$: differential equation $\rightarrow$ **extra degree of freedom**

De Sitter condition (vacuum, $R = \text{const}$):

$$Rf_R(R) - 2f(R) = 0 \quad \Rightarrow \quad \Lambda_{\text{eff}} = \frac{R_{\text{dS}}}{4}$$

The Scalon and Chameleon Mechanism

Scalar-tensor equivalence:

$$\phi \equiv f_R(R) \quad (\text{scalon field})$$

Effective scalon mass:

$$m_\phi^2(R) = \frac{f_R - Rf_{RR}}{3f_{RR}}$$

Yukawa correction to Newton potential:

$$\Phi(r) \simeq -\frac{GM}{f_R r} \left[1 + \frac{1}{3} e^{-m_\phi r} \right]$$

Chameleon Screening

High density $\rightarrow$ large $m_\phi \rightarrow$ short range $\rightarrow$ GR recovered

Viability Conditions

Theoretical Constraints:

- ① **No-ghost:** $f_R > 0$
(positive gravitational coupling)
- ② **Dolgov-Kawasaki:** $f_{RR} > 0$
(no tachyonic instability)
- ③ **de Sitter stability:**
$$0 < \frac{R_{\text{dS}} f_{RR}(R_{\text{dS}})}{f_R(R_{\text{dS}})} < 1$$

Observational Constraints:

- ① **GR recovery at high R :**

$$f(R) \rightarrow R, \quad f_R \rightarrow 1$$

- ② **Standard matter era:**

$$|f_R - 1| \ll 1 \text{ during structure formation}$$

- ③ **Solar System tests:**

$$|f_{R,0} - 1| \lesssim 10^{-6}$$

All conditions must be satisfied simultaneously!

UV regime screened Model Definition

Screened Logarithmic $f(R)$ Model

$$f(R) = R + \alpha R \frac{\ln(R/R_0)}{1 + (R/R_0)^n}$$

Model Parameters:

- $R_0 > 0$: characteristic curvature scale
- $\alpha > 0$: dimensionless coupling constant
- $n \geq 1$: screening efficiency exponent

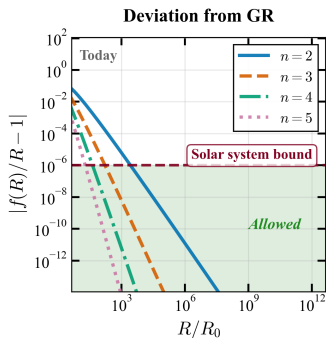
Design Philosophy:

- 1 Logarithm $\rightarrow$ scale-dependent modification
- 2 Screening factor $1 + (R/R_0)^n \rightarrow$ automatic GR recovery at high R
- 3 Overall R factor $\rightarrow$ Minkowski remains a solution
- 4 Only 3 free parameters $\rightarrow$ tractable phenomenology

High-Curvature Screening: Deviation from GR

Defining $x \equiv R/R_0$, the fractional deviation from GR:

$$\frac{f(R) - R}{R} = \alpha \frac{\ln x}{1 + x^n} \xrightarrow{x \gg 1} \alpha \frac{\ln x}{x^n} \rightarrow 0$$

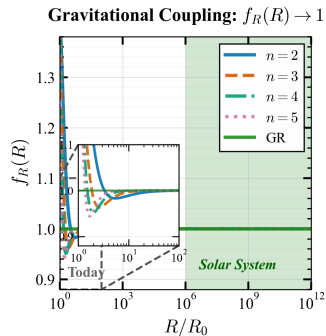


Deviations suppressed below Solar System bound

Effective gravitational coupling:

$$G_{\text{eff}} \propto 1/f_R$$

At high curvature: $f_R - 1 = \mathcal{O}(\alpha \frac{\ln x}{x^n}) \rightarrow 0$

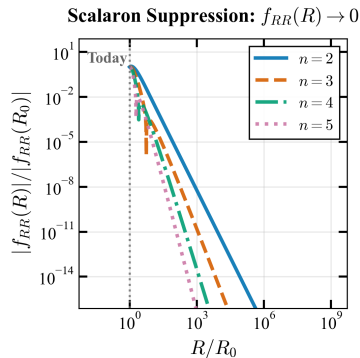


$f_R(R) \rightarrow 1$ (GR value) at high curvature. No antigravity regime ($f_R > 0$ always).

Screening Implications

What the screening ensures:

- ✓ **Early Universe** ($R \gg R_0$):
GR preserved, standard cosmology
- ✓ **BBN & CMB physics:**
Unaffected by modifications
- ✓ **Solar System:**
scalaron suppressed
- ✓ **Laboratory tests:**
GR recovered in high-density



Scalaron decouples ($f_{RR} \rightarrow 0$) at high curvature

Effective IR model and De Sitter Solution

Effective late-time model (when $R \approx R_0$):

$$f(R) \simeq R + \alpha R \ln\left(\frac{R}{R_0}\right)$$

De Sitter condition $Rf_R - 2f = 0$ gives:

$$R \left[1 - \alpha + \alpha \ln\left(\frac{R}{R_0}\right) \right] = 0$$

Solutions

Trivial: $R = 0$
(Minkowski spacetime)

Non-trivial:

$$R_{\text{dS}} = R_0 e^{1-1/\alpha}$$

(de Sitter spacetime)

For $\alpha > 0$: $R_{\text{dS}} > 0$ ✓ For $0 < \alpha < 1$: $R_{\text{dS}} < R_0$ (late-time acceleration)

Effective Cosmological Constant

From the de Sitter solution: Requiring $R_{\text{dS}} = 4\Lambda$ yields the matching condition

$$R_0 e^{1-1/\alpha} = 4\Lambda, \quad (1)$$

which can be inverted to give

$$\alpha = \frac{1}{1 - \ln(4\Lambda/R_0)}. \quad (2)$$

The curvature scale R_0 is,

$$R_0 \equiv R(t_0) = 3H_0^2(\Omega_{m0} + 4\Omega_{\Lambda0}), \quad (3)$$

while $\Lambda = 3H_0^2\Omega_{\Lambda0}$. Using Planck 2018 values ($\Omega_{m0} \simeq 0.315$, $\Omega_{\Lambda0} \simeq 0.685$), one finds

$$\frac{4\Lambda}{R_0} = \frac{4\Omega_{\Lambda0}}{\Omega_{m0} + 4\Omega_{\Lambda0}} \simeq 0.9, \quad \alpha \simeq 0.9. \quad (4)$$

Stability of the De Sitter Solution

Scalaron mass at R_{dS} :

$$m_\phi^2(R_{\text{dS}}) = \frac{f_R(R_{\text{dS}}) - R_{\text{dS}} f_{RR}(R_{\text{dS}})}{3f_{RR}(R_{\text{dS}})}$$

Using $f_R = 1 + \alpha + \alpha \ln(R/R_0)$ and $f_{RR} = \alpha/R$:

$$m_\phi^2(R_{\text{dS}}) = \frac{R_{\text{dS}}}{3} > 0 \quad \checkmark$$

Perturbation equation:

$$\delta\ddot{R} + 3H_{\text{dS}}\delta\dot{R} + m_\phi^2\delta R = 0$$

- Damped harmonic oscillator
- Perturbations decay exponentially
- **De Sitter is a stable attractor**

$$m_\phi(R_{\text{dS}}) = \frac{H_{\text{dS}}}{\sqrt{3}} \sim H_0$$

Modified Friedmann Equations

FRW metric (flat, $k = 0$):

$$ds^2 = -dt^2 + a^2(t) [dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)]$$

Ricci scalar: $R = 12H^2 + 6\dot{H}$

Modified Friedmann equations for the logarithmic model:

$$3H^2 \left[1 + \alpha + \alpha \ln \left(\frac{R}{R_0} \right) \right] = \kappa^2 \rho + \frac{\alpha R}{2} \left[1 + \ln \left(\frac{R}{R_0} \right) \right] - \frac{3\alpha H \dot{R}}{R}$$

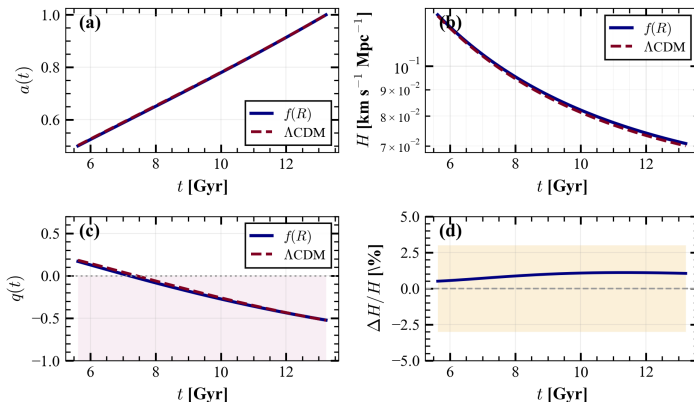
$$-2\dot{H} \left[1 + \alpha + \alpha \ln \left(\frac{R}{R_0} \right) \right] = \kappa^2(\rho + p) + \frac{\alpha \ddot{R}}{R} - \frac{\alpha \dot{R}^2}{R^2} - \frac{\alpha H \dot{R}}{R}$$

Closed system with continuity equation: $\dot{\rho} + 3H(\rho + p) = 0$

Numerical Results: Cosmological Evolution

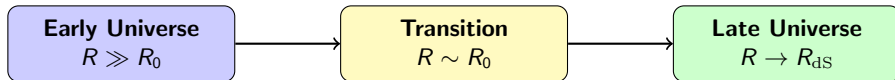
Numerical integration with Planck 2018 parameters:

$\Omega_m = 0.315$, $\Omega_\Lambda = 0.685$, $H_0 = 67.4$ km/s/Mpc, **Best-fit value:** $\alpha \simeq 0.967$



Comparison $f(R)$ vs Λ CDM: (a) scale factor, (b) Hubble parameter, (c) deceleration parameter, (d) relative deviation $< 1\%$

Physical Picture



Screening active
GR behavior
Radiation/Matter eras

Modification
becomes relevant
 w_{eff} deviates

de Sitter attractor
Geometric acceleration
No explicit Λ

→ time

Theoretical Investigations:

- Perturbation theory & structure growth
- Matter power spectrum predictions
- CMB anisotropy signatures
- Gravitational wave propagation

Observational Confrontation:

- SNIa, BAO, CMB constraints on (α, R_0, n)
- Solar System tests (perihelion, Shapiro delay)
- Galaxy/cluster screening analysis
- Multi-messenger GW constraints

Thank you!

Questions?

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